

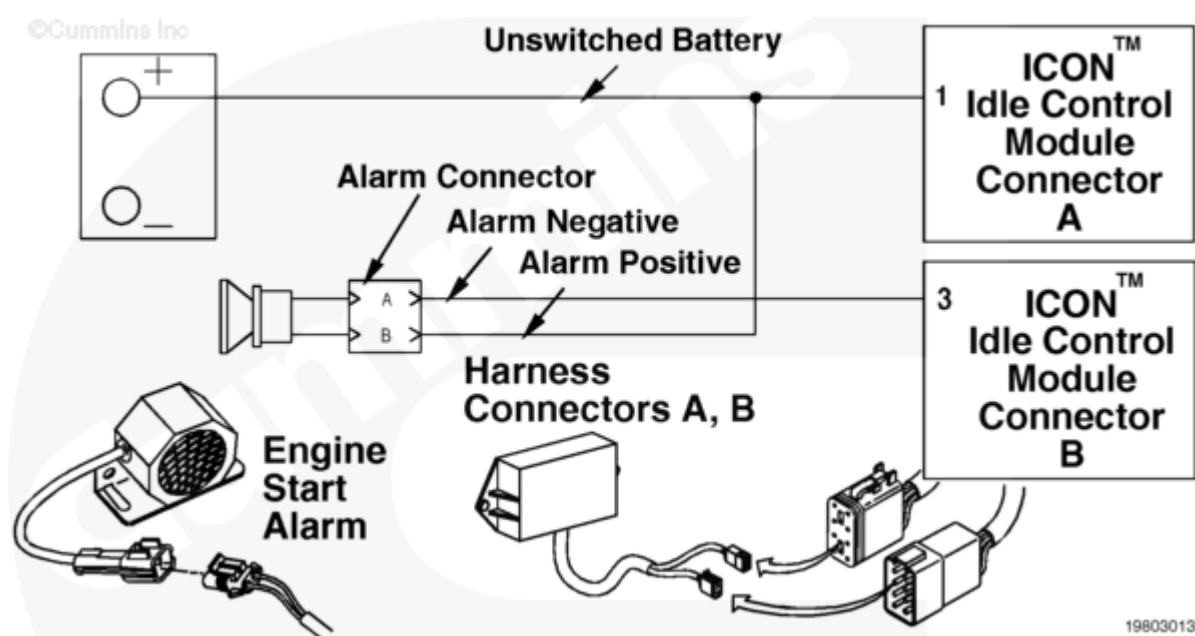
Fault Code: 589 (Aftermarket and OEM)

Auto Start Alarm Circuit - Voltage Below Normal or Shorted to Low Source

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 589 PID(P): SPN: FMI: Lamp: SRT:	Auto Start Alarm Circuit - Voltage Below Normal or Shorted to Low Source. Less than 6 VDC detected at the alarm circuit when high voltage was expected by the ICON™ idle control module.	The ICON™ system will be disabled. Only mandatory shutdown will be enabled. The alarm will possibly sound continuously if there is a short circuit to ground. Engine will start normally.



Circuit Description

The alarm circuit turns on the audible alarm to warn of an impending engine start. The above circuit diagram can vary, such as connector or pins, depending on the vehicle make or model. OEM installations can possibly provide the harnessing between the idle control module and other ICON™ devices.

Component Location

The engine start alarm is typically mounted on the vehicle bulkhead on the intake side of the engine.

The ICON™ module can be located in a different location depending on the vehicle application.

Shoptalk

This fault typically indicates an open circuit between the alarm coil (pin A and B), or from pin B of the alarm to battery voltage, or from pin A of the alarm to pin 3 of the ICON™ idle control module. Another typical cause for this fault is a short circuit from pin A or B of the alarm, or pin 3 of the ICON™ idle control module to ground. The alarm can possibly sound continuously if there is a short circuit to ground. The alarm can sound for 14 seconds before an engine start. The alarm is actuated by applying a ground from pin 3 of the ICON™ idle control module B connector to pin A of the alarm connector.

The ICON™ system can display **only** the present active fault code. If more than one fault code is active at the same time, the ICON™ system flashes out the highest priority fault. After the fault has been corrected then the next active fault will be flashed out.

Note: The ICON™ electronic service tool can display more than one active and or inactive fault codes at the same time.

Warnings and Cautions

CAUTION

To reduce the possibility of damaging the new ICON™ idle control module, all other active fault codes must be investigated prior to replacing the ICON™ idle control module.

CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement:

Part Number 3822758 - male Deutsch/AMP/Metri-Pack test lead

Part Number 3822917 - female Deutsch/AMP/Metri-Pack test lead.

Troubleshooting Summary

STEPS

SPECIFICATIONS

STEP 1.

Check the engine start alarm operation.

STEPS		SPECIFICATIONS
	STEP 1A. Check the alarm status.	Alarm not on
	STEP 1B. Check the ICON™ idle control module for a short to ground.	Alarm not on
	STEP 1C. Test the engine start alarm.	Alarm sounds
	STEP 1D. Check the ICON™ idle control module operation.	Alarm sounds and fault code inactive
STEP 2.	Check the engine start alarm.	
	STEP 2A. Inspect the engine start alarm connector pins.	No damaged pins
	STEP 2A-1. Inspect the engine harness start alarm connector pins.	No damaged pins
	STEP 2B. Check the engine start alarm resistance.	800 to 1200 ohms
STEP 3.	Check the ICON™ engine or OEM harness.	
	STEP 3A. Inspect the ICON™ engine harness and ICON™ idle control module connector pins.	No damaged pins
	STEP 3B. Check for a short circuit from pin to pin.	More than 100k ohms
	STEP 3C. Check for a short circuit to ground.	More than 100k ohms
STEP 4.	Clear the fault code.	
	STEP 4A. Disable the fault code.	Fault Code 589 cleared

STEP 1. Check the engine start alarm operation.

STEP 1A. Check the engine start alarm status.

Conditions :

- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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	Alarm not on	1C
	Alarm is on	1B

STEP 1B. Check the ICON™ idle control module for a short to ground.

Conditions :

- Turn keyswitch ON.

Action	Specification/Repair	Next Step
	Alarm not on Replace the ICON™ idle control module. Refer to Procedure 019-358 (/qs3/pubsys2/xml/en/procedures/97/97-019-358.html).	Repair Complete
	Alarm on	2A

STEP 1C. Test the engine start alarm.

Conditions :

- Turn keyswitch OFF.
- Connect the ICON™ electronic service tool.

Action	Specification/Repair	Next Step
	Alarm sounds	1D
		2A

STEP 1D. Check the ICON™ idle control module operation.

Conditions :

- Turn keyswitch ON.
- Connect the ICON™ electronic service tool.
- Start the engine.

Action	Specification/Repair	Next Step
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	Alarm sounds and fault code inactive Refer to Inactive or Intermittent Fault Codes, Procedure 019-362 (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html).	4A
	Replace the ICON™ idle control module. Refer to Procedure 019-358 (/qs3/pubsys2/xml/en/procedures/97/97-019-358.html).	Repair Complete

STEP 2. Check the engine start alarm.

STEP 2A. Inspect the engine start alarm connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine start alarm connector from the ICON™ engine harness.

Action	Specification/Repair	Next Step

• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Wire insulation damage • Moisture in or on the connector • Missing or damaged connector seals • Connector shell broken • Dirt or debris in or on the connector pins. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	2A-1
	Repair the connector pins. • Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Repair the connector pins. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html) or 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html).	4A

STEP 2A-1. Inspect the engine harness start alarm connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ start alarm from the ICON™ engine harness.

Action	Specification/Repair	Next Step

• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Wire insulation damage • Moisture in or on the connector • Missing or damaged connector seals • Connector shell broken • Dirt or debris in or on the connector pins. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	2B
	Repair the damaged pins. • Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Repair the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html). • Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html). • Repair or replace the OEM wiring harness as necessary.	4A

STEP 2B. Check the engine start alarm resistance.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine start alarm connector from the ICON™ engine harness.

Action	Specification/Repair	Next Step
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Measure the resistance from pin A of the engine start alarm connector to pin B of the connector. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	800 to 1200 ohms	3A
	Replace the engine start alarm. Refer to Procedure 019-293 (/qs3/pubsys2/xml/en/procedures/97/97-019-293.html).	4A

STEP 3. Check the ICON™ engine or OEM harness.

STEP 3A. Inspect the ICON™ engine harness and ICON™ idle control module connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ idle control module A and B connectors from the ICON™ idle control module.

Action	Specification/Repair	Next Step

• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Wire insulation damage • Moisture in or on the connector • Missing or damaged connector seals • Connector shell broken • Dirt or debris in or on the connector pins. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	3B
	Repair the damaged pins. • Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).	4A

STEP 3B. Check for a short circuit from pin to pin.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ idle control module B connector from the ICON™ idle control module.
- Disconnect the engine start alarm connector from the ICON™ engine harness.

Action	Specification/Repair	Next Step

Measure the resistance from pin 3 of the ICON™ idle control module B harness connector to all other pins in the connector. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms	3C
	Repair or replace the ICON™ engine harness. Repair the ICON™ engine harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html) or 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html).	4A

STEP 3C. Check for a short circuit to ground.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine start alarm connector from the ICON™ engine harness.
- Disconnect the ICON™ idle control module B connector from the ICON™ idle control module.

Action	Specification/Repair	Next Step
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Measure the resistance from pin 3 of the ICON™ idle control module B harness connector to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms Replace the ICON™ idle control module. Refer to Procedure 019-358 (/qs3/pubsys2/xml/en/procedures/97/97-019-358.html).	Repair Complete.
	- Repair the ICON™ engine harness. Refer to Procedure 019-197 (/qs3/pubsys2/xml/en/procedures/99/99-019-197.html), 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html), or 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html). - Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html). - Repair or replace the OEM wiring harness as necessary.	4A

STEP 4. Clear the fault code.

STEP 4A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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• Cycle the keyswitch to verify the fault code is inactive.	Fault Code 589 cleared	Repair complete
	Troubleshoot any remaining active fault codes.	Appropriate troubleshooting charts